

# Rizee Foundation

# Product Requirements Document (PRD)

## Iteration 1 – Product Foundation

VERSION 1.0

DOCUMENT OWNER: PRODUCT TEAM

STATUS: DRAFT

| Version | Date    | Author       | Changes                          |
|---------|---------|--------------|----------------------------------|
| 1.0     | Current | Product Team | Initial PRD — Product Foundation |

**Purpose:** This PRD provides the functional direction required for product, design and engineering teams to build a scalable and engaging platform. The Rizee Foundation module is designed as a complete AI-powered learning ecosystem rather than a conventional LMS. Every module—from learning and assessments to Smart Revision and Goal Planning—works together to create a personalized academic journey.

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# Executive Summary & Product Vision

Rizee Foundation is an AI-powered learning ecosystem that transforms traditional study material into a guided learning experience. The platform combines AI-generated notes, adaptive practice, intelligent assessments, analytics, rewards, Smart Revision and Goal Planning to help students build conceptual clarity and maintain consistent preparation. Every feature is designed to answer one question for the learner: **'What should I study next, and why?'**

The vision is to make Rizee Foundation the student's everyday academic companion. Instead of forcing students to decide what to study, the platform proactively recommends learning tasks based on syllabus progress, assessment performance and revision history.

| Objective    | Description                                                                |
|--------------|----------------------------------------------------------------------------|
| Mission      | Deliver a personalized AI-driven learning experience.                      |
| Primary Goal | Improve learning consistency and conceptual mastery.                       |
| Success      | Higher engagement, stronger retention and measurable academic improvement. |

# Problems Being Solved & Target Audience

## Problems Being Solved

- Students struggle to decide what to study next.
- Revision is inconsistent.
- Content, tests and analytics are disconnected.
- Parents and teachers lack actionable insights.
- Study planning is manual and inefficient.

## Target Audience

The module serves Foundation students, parents, teachers and administrators. Students use the platform for daily learning while teachers and parents monitor progress and administrators manage academic content.

## User Personas

### Student

Reads notes, practices, takes tests, revises and tracks progress.

### Parent

Monitors consistency, performance and improvement.

### Teacher

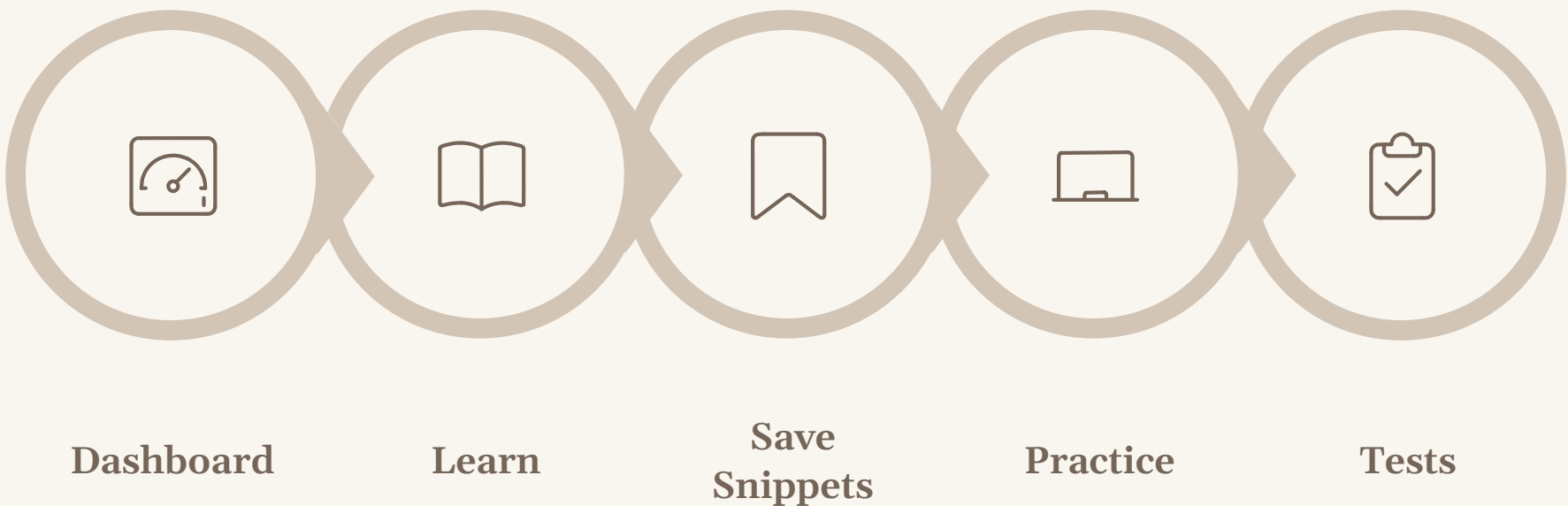
Assigns learning activities and reviews analytics.

### Administrator

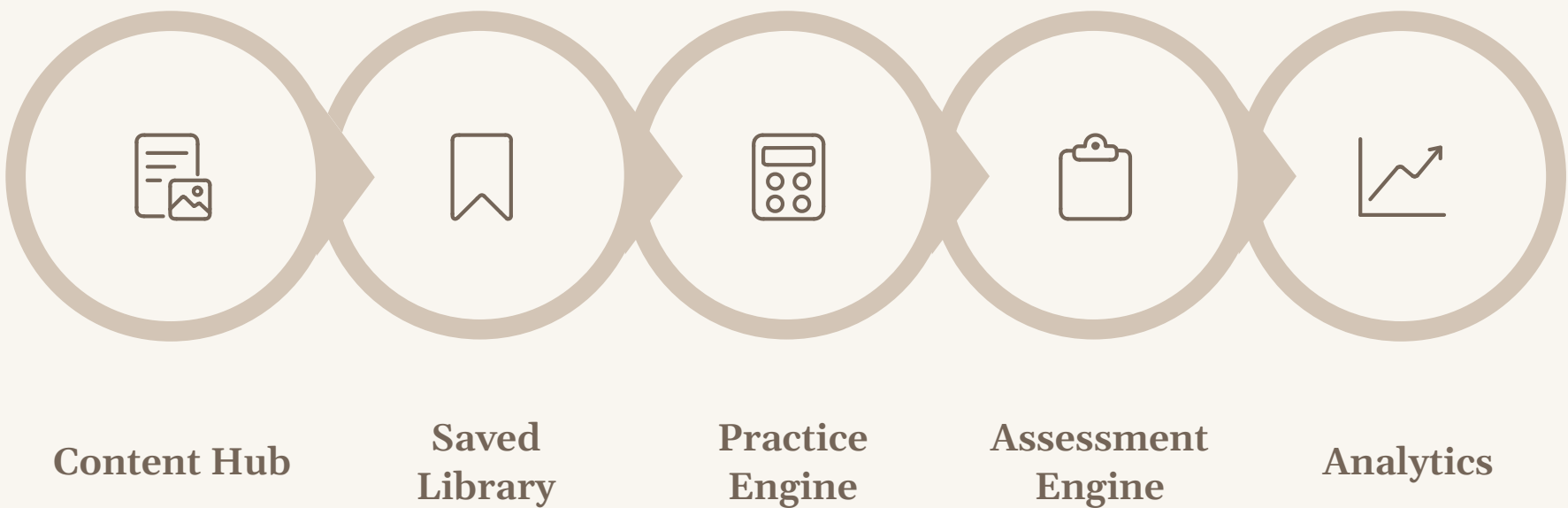
Manages syllabus, content, users and AI-generated material.

# End-to-End Student Journey & Platform Architecture

The student begins at the Dashboard where today's learning goals and AI recommendations are displayed. They continue learning through interactive notes, save snippets, annotate important concepts, complete adaptive practice, attempt assessments and receive personalized analytics. Weak topics automatically enter the Smart Revision queue while the Goal Planner updates the daily study schedule.



Every module contributes learning data that improves the overall experience. The Content Hub powers learning, assessments measure understanding, analytics identify gaps, Smart Revision improves retention and the Goal Planner converts insights into daily action.





# Dashboard & Learn Module

The Dashboard and Learn Module together form the primary workspace of every student. The dashboard acts as an intelligent command centre, while the Learn Module provides an interactive reading experience with AI-generated notes, annotations and personalized learning tools. Every interaction feeds the analytics engine and helps the AI recommend the next best learning activity.

## Dashboard

Immediately after login, students arrive at the Dashboard. Instead of presenting every available feature, the dashboard prioritizes what the student should do today. It minimizes decision fatigue by surfacing only the most relevant actions.

| Widget            | Purpose                              | Behaviour                             |
|-------------------|--------------------------------------|---------------------------------------|
| Daily Goal        | Shows today’s study target           | Updated automatically by Goal Planner |
| Continue Learning | Resume last chapter                  | Appears only if progress <100%        |
| Daily Challenge   | Launch 5-question reward quiz        | Resets every day                      |
| AI Recommendation | Suggest next chapter or revision     | Generated from performance            |
| Weekly Progress   | Visual learning trend                | Updates after every activity          |
| Saved Content     | Quick access to snippets & bookmarks | Shows recent additions                |

 **Why this Matters:** Students often waste valuable time deciding where to begin. The dashboard removes this uncertainty by guiding them directly toward the highest-priority learning activity.

## Learn Module

The Learn Module transforms static study material into an interactive digital experience. Notes are organized as Class → Subject → Chapter → Topic. Each topic contains concise AI-generated explanations, worked examples, formulae, tricks, common mistakes and a quick summary before linking directly to practice questions.

### Reading Experience

- Clean distraction-free reading interface with adjustable font size.
- Estimated reading time and completion progress.
- Inline concept tags and difficulty indicators.
- Dark mode for extended reading sessions.
- Automatic progress synchronization across devices.

## Interactive Learning Features

| Feature        | Description                      | Student Benefit                     |
|----------------|----------------------------------|-------------------------------------|
| Highlight      | Mark important text              | Quick revision                      |
| Save Snippets  | Save selected paragraphs         | Build personalized revision library |
| Bookmarks      | Save entire notes                | Fast access later                   |
| Doodle         | Draw and annotate over notes     | Improves retention                  |
| Personal Notes | Attach custom notes to any topic | Capture individual understanding    |
| Search         | Search across notes              | Locate concepts instantly           |

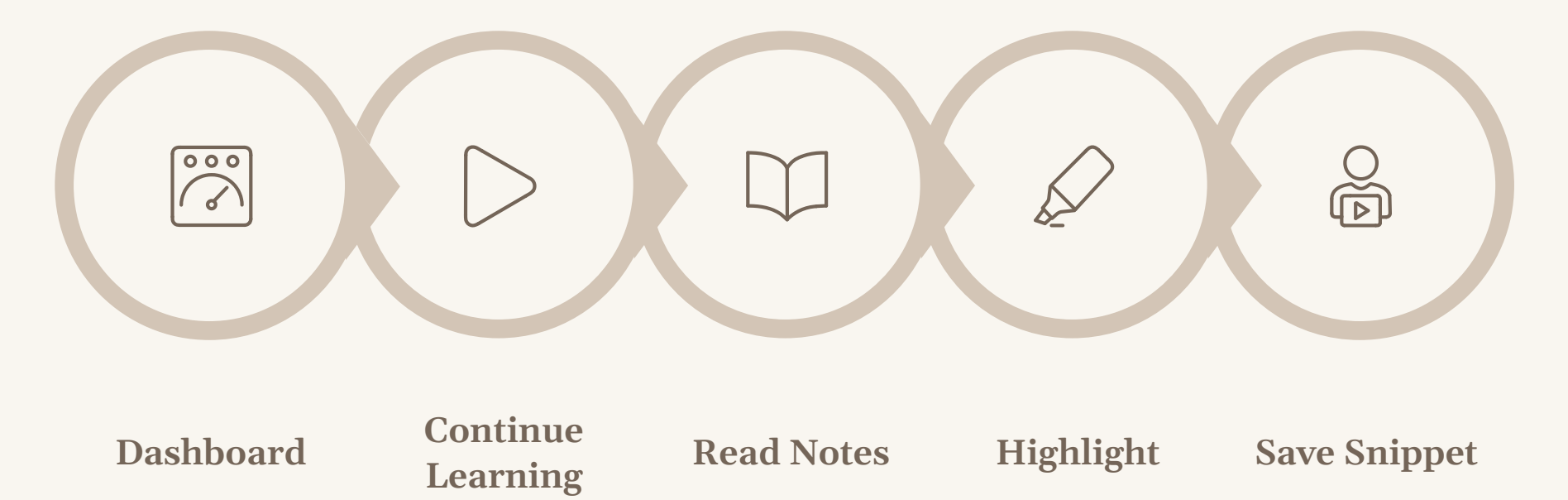
## Saved Library

The Saved Library consolidates all student-created learning material into one place. It stores snippets, bookmarks, handwritten annotations and personal notes. Each saved item remains linked to its original chapter, enabling students to revisit the full context during revision.

## AI Recommendations

The Learn Module continuously evaluates reading behaviour. If a student spends excessive time on a topic, skips important concepts or repeatedly performs poorly in related assessments, the recommendation engine automatically suggests additional notes, practice questions or revision packs.

## User Flow



 **Developer Notes:** All annotations should auto-save. Snippets must preserve chapter, topic and timestamp metadata. Dashboard widgets should load independently to reduce perceived loading time and improve responsiveness.

# Practice & Assessment Engine

The Practice and Assessment Engine converts learning into measurable outcomes. After consuming content in the Learn Module, students reinforce concepts through structured practice and examinations. Every attempt generates data that feeds the Analytics Dashboard, Smart Revision Engine and Goal Planner, ensuring that future recommendations become increasingly personalized.

## Learning Flow



## Practice Modes

| Mode              | Purpose                       | Typical Size         | AI Behaviour                     |
|-------------------|-------------------------------|----------------------|----------------------------------|
| Topic Practice    | Reinforce a single concept    | 10–15 Questions      | Focus on current topic           |
| Chapter Practice  | Consolidate an entire chapter | 25–40 Questions      | Mix easy, medium and hard        |
| Mixed Practice    | Revision across chapters      | 20 Questions         | Weighted to weak chapters        |
| Timed Sprint      | Improve speed                 | 10 Questions / 5 min | Time pressure                    |
| Adaptive Practice | Personalized learning         | 15–20 Questions      | Difficulty adjusts continuously  |
| Mistake Practice  | Retry previous errors         | Variable             | Only incorrect/skipped questions |

**Why this Matters:** Students learn best when practice is purposeful. Rather than presenting random questions, the engine selects questions that reinforce concepts, identify gaps and build confidence before formal assessments.

## Assessment Engine

Assessments measure conceptual understanding at different stages of the learning journey. Each test type has a distinct purpose and contributes unique insights to the student’s learning profile.

| Test            | Purpose                   | Frequency   | Outcome            |
|-----------------|---------------------------|-------------|--------------------|
| Daily Challenge | Daily habit building      | Daily       | Rewards & streak   |
| Chapter Test    | End of chapter evaluation | Per chapter | Mastery score      |
| Weekly Test     | Weekly revision           | Weekly      | Progress trend     |
| Monthly Test    | Long-term retention       | Monthly     | Performance review |
| Grand Test      | Full syllabus simulation  | Periodic    | Exam readiness     |
| Error Test      | Correct misconceptions    | As needed   | Revision queue     |
| Olympiad Test   | Higher-order thinking     | Optional    | Advanced learning  |
| Foundation Test | IIT foundation concepts   | Optional    | Concept depth      |

## Result Analysis

Immediately after submission, the platform generates a comprehensive report instead of only displaying marks.

- Overall score and percentage
- Accuracy by chapter and topic
- Time spent per question
- Difficulty-wise performance
- Questions answered correctly, incorrectly and skipped
- Detailed explanations and correct solutions
- AI-generated recommendations for improvement

## Error Detection & AI Response

Incorrect answers are categorized into conceptual errors, calculation mistakes, careless mistakes and time-management issues. Depending on the category, the system recommends targeted revision notes, additional practice or inclusion in the Smart Revision queue.

## Functional Requirements

| Requirement ID | Requirement                                               | Priority |
|----------------|-----------------------------------------------------------|----------|
| FR-16          | Support adaptive practice with dynamic difficulty         | High     |
| FR-17          | Generate detailed result analytics after every assessment | High     |
| FR-18          | Automatically create Error Tests from previous mistakes   | High     |
| FR-19          | Link assessment results to Smart Revision                 | High     |
| FR-20          | Recommend next learning activity based on performance     | Medium   |

**Developer Notes:** Question banks must be tagged by class, subject, chapter, topic, subtopic, difficulty, cognitive level and learning objective. The assessment engine should support randomized questions and options while preventing excessive repetition across attempts.

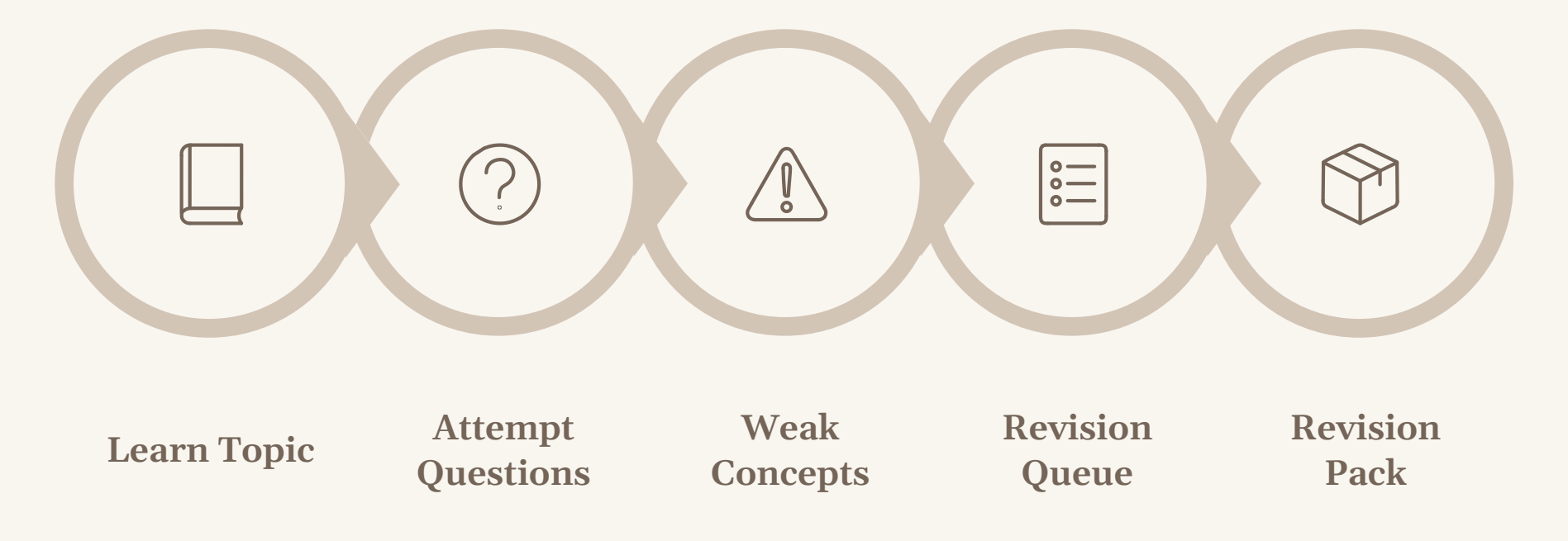
# Smart Revision Engine & Goal Planner

The Smart Revision Engine and Goal Planner form the intelligence layer of the Rizee Foundation platform. While the Learn and Assessment modules capture how a student studies, these two modules determine what the student should do next. They transform raw learning data into actionable daily plans that continuously adapt to the learner’s progress.

## Smart Revision Engine

Traditional revision depends entirely on the student’s memory and planning. The Smart Revision Engine removes this burden by automatically creating revision schedules using learning history, assessment performance and spaced repetition principles. The objective is to ensure that concepts are revised just before they are likely to be forgotten.

### Revision Workflow



### Data Sources Used for Revision

- Incorrectly answered questions
- Skipped questions
- Low-confidence topics
- Saved snippets
- Bookmarked notes
- Doodled or annotated content
- Time since last revision
- Upcoming examinations

### Revision Priority Logic

| Priority | Condition                  | Action                   |
|----------|----------------------------|--------------------------|
| Critical | Repeated incorrect answers | Immediate revision       |
| High     | Low accuracy (<60%)        | Schedule within 24 hours |
| Medium   | No revision for 7+ days    | Add to weekly revision   |
| Low      | Concept already mastered   | Monthly refresher        |

### Revision Packs

Instead of asking students to revise an entire chapter, the platform creates compact Revision Packs that can be completed in 10–15 minutes. Each pack includes important snippets, formulae, previously incorrect questions, AI-generated summaries and a short mastery quiz.

## Goal Planner

The Goal Planner converts long-term academic goals into manageable daily activities. Students begin by selecting their target examination, expected score or rank, available study hours and exam date. Using this information, the AI prepares a realistic study schedule that balances learning, practice and revision.

### Planner Inputs

- Target examination
- Target score / rank
- Exam date
- Daily study hours
- Current syllabus completion
- Historical performance

### Daily Study Plan

Each day, the planner generates three to five clear tasks. These may include reading a chapter, completing practice questions, attempting a test or finishing a Revision Pack. If the student misses tasks, the planner automatically redistributes the workload across future days without overwhelming the learner.

### Progress Tracking

| Metric              | Purpose                    | Student Benefit        |
|---------------------|----------------------------|------------------------|
| Goal Completion     | Track completed tasks      | Motivation             |
| Study Consistency   | Monitor daily learning     | Habit building         |
| Predicted Readiness | Estimate exam preparedness | Confidence             |
| Missed Tasks        | Identify pending work      | Automatic rescheduling |

## Functional Requirements

| Requirement ID | Requirement                            | Priority |
|----------------|----------------------------------------|----------|
| FR-21          | Automatically generate revision queues | High     |
| FR-22          | Create AI-powered Revision Packs       | High     |
| FR-23          | Generate dynamic daily study plans     | High     |
| FR-24          | Automatically reschedule missed tasks  | High     |
| FR-25          | Predict exam readiness using analytics | Medium   |

- Why this Matters:** Students usually know where they are struggling but often do not know how to recover. The Smart Revision Engine and Goal Planner bridge this gap by automatically converting performance data into structured learning actions. Rather than simply highlighting weaknesses, the platform actively guides students toward improvement.
- Developer Notes:** Revision scheduling should be configurable to support different educational philosophies and examination patterns. The planner should expose APIs that allow future AI models to optimise schedules based on behavioural insights, engagement and historical success rates.



# Analytics, Syllabus Intelligence & AI Recommendation Engine

The Analytics, Syllabus Intelligence and AI Recommendation modules transform raw student activity into meaningful insights and actionable guidance. Rather than displaying only scores, these modules explain why a student is performing a certain way and recommend the next steps required to improve outcomes.

## Student Analytics Dashboard

The Analytics Dashboard serves as the student’s academic control centre. It aggregates data from learning sessions, practice, assessments, Smart Revision and the Goal Planner to provide a holistic view of progress.

| Widget              | Purpose                        | Insight                |
|---------------------|--------------------------------|------------------------|
| Overall Performance | Track cumulative score         | Academic progress      |
| Subject Mastery     | Measure subject understanding  | Strong & weak subjects |
| Topic Mastery       | Identify concept gaps          | Targeted revision      |
| Study Consistency   | Track daily study habits       | Habit formation        |
| Reading Analytics   | Monitor learning time          | Engagement             |
| Goal Progress       | Measure planner completion     | Target achievement     |
| Revision Status     | Pending vs completed revisions | Retention level        |

## Parent & Teacher Insights

Separate dashboards provide stakeholders with role-specific information. Parents receive simplified reports focusing on consistency and improvement, while teachers receive class-level analytics to identify students requiring intervention.

- Weekly performance summaries
- Attendance to learning plans
- Top strengths and improvement areas
- Batch-wise comparison for teachers
- Student engagement reports

## Syllabus Intelligence

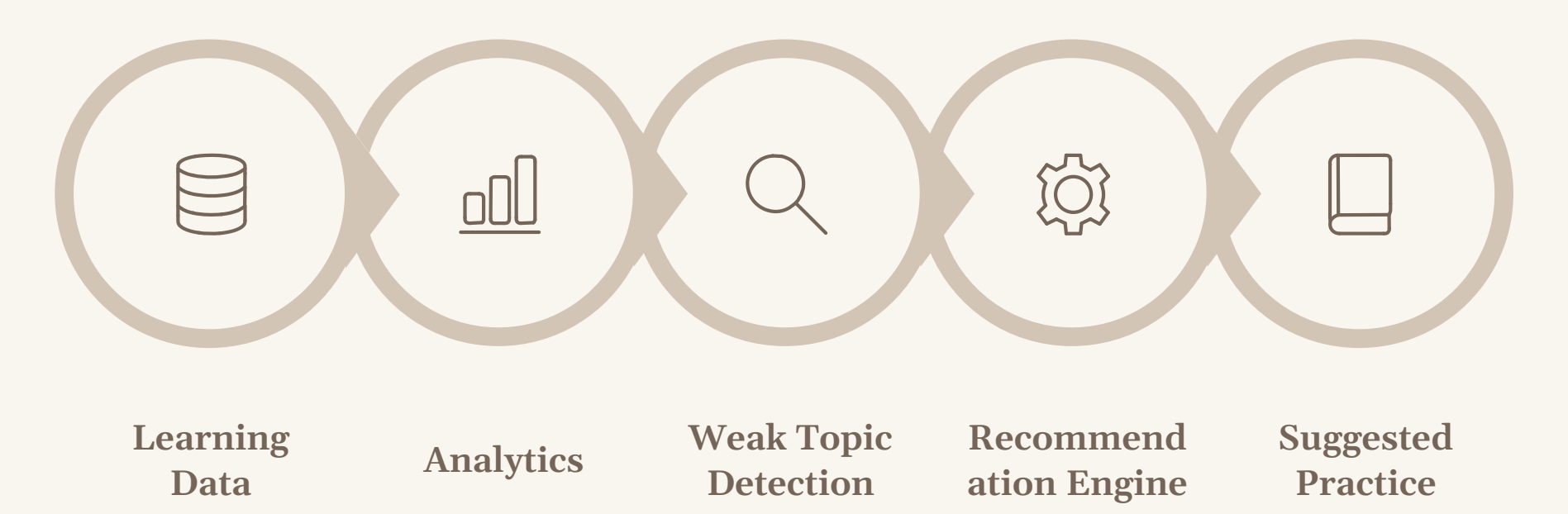
Instead of displaying a static syllabus, the platform presents an interactive roadmap. Every chapter is enriched with metadata that helps students understand its importance and recommended order of study.

| Attribute      | Description              | Displayed To | Purpose           |
|----------------|--------------------------|--------------|-------------------|
| Importance     | High / Medium / Low      | Student      | Prioritize study  |
| Prerequisites  | Topics to complete first | Student      | Logical learning  |
| Estimated Time | Hours required           | Student      | Planning          |
| Weightage      | Exam relevance           | Student      | Focus preparation |
| Completion     | Covered / In Progress    | All          | Track syllabus    |

## AI Recommendation Engine

The AI Recommendation Engine acts as the academic mentor of the platform. Using learning behaviour, assessment results and revision history, it continuously recommends the next best activity for every learner.

### Recommendation Workflow



### Recommendation Types

- Next chapter to study
- Concepts requiring immediate revision
- Recommended practice questions
- Suggested tests
- Daily study goals
- Revision packs before examinations
- Motivational reminders based on consistency

## Institutional Analytics

Institution administrators require macro-level insights to evaluate academic effectiveness across schools, batches and programmes.

| Metric              | Purpose             | Business Value          |
|---------------------|---------------------|-------------------------|
| Active Students     | Monitor adoption    | Platform growth         |
| Average Completion  | Track engagement    | Content effectiveness   |
| Assessment Scores   | Academic quality    | Institution performance |
| Revision Compliance | Learning discipline | Retention improvement   |
| Teacher Activity    | Usage monitoring    | Operational insights    |

## Functional Requirements

| Requirement ID | Requirement                         | Priority |
|----------------|-------------------------------------|----------|
| FR-26          | Provide student analytics dashboard | High     |
| FR-27          | Provide parent & teacher dashboards | High     |
| FR-28          | Display syllabus importance mapping | High     |
| FR-29          | Generate AI recommendations         | High     |
| FR-30          | Provide institutional analytics     | Medium   |

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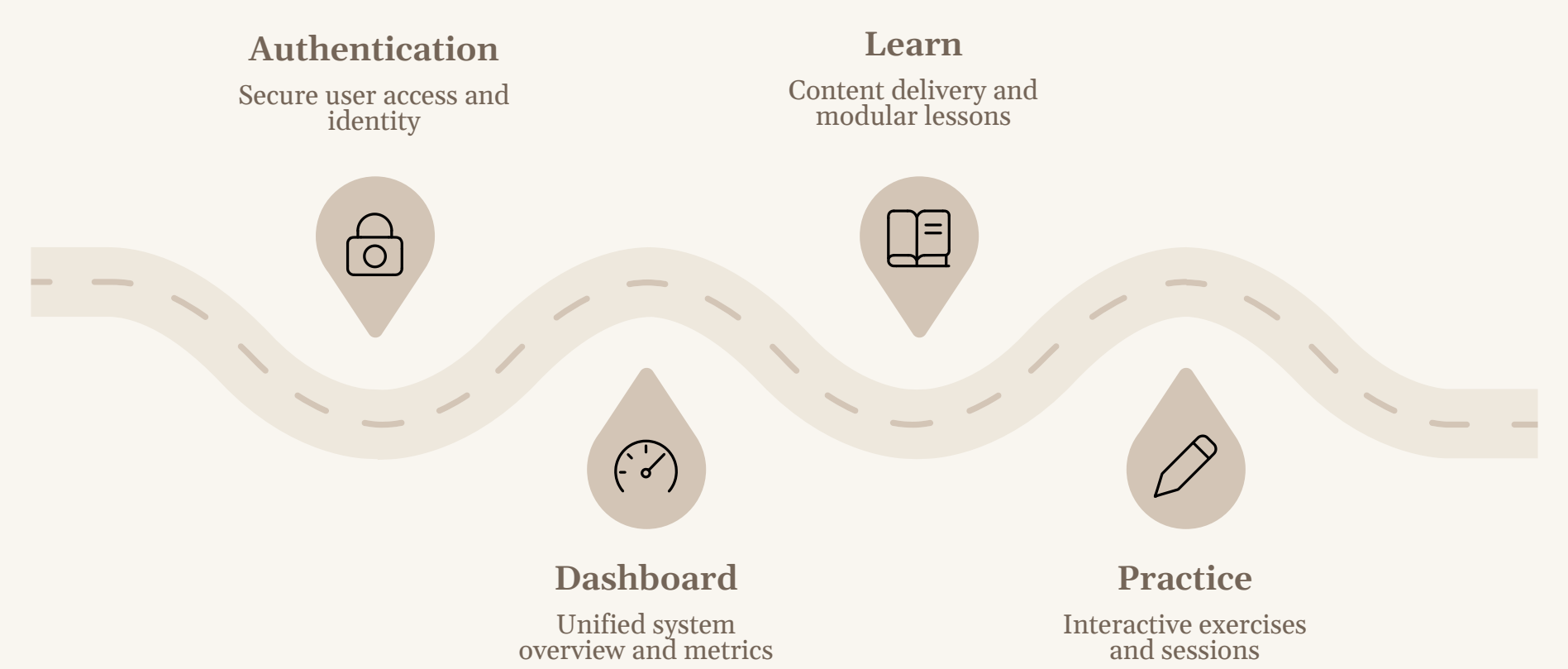
**Business Impact:** These modules convert platform data into actionable intelligence. Students receive personalized guidance, parents gain transparency, teachers identify intervention opportunities and institutions obtain measurable academic insights that support data-driven decision making.
- ☐

**Developer Notes:** The analytics service should be event-driven so every meaningful student interaction updates dashboards asynchronously. Recommendation models should remain configurable to support future AI enhancements without changes to the core platform.

# Technical Blueprint, Roadmap & Implementation

## Technical Architecture

The Foundation platform follows a modular architecture where every major capability is implemented as an independent service. This approach allows individual modules such as Learning, Assessments, Analytics and AI Recommendations to evolve without impacting the rest of the platform.



## Core Services

| Service                | Responsibility                          | Dependencies          |
|------------------------|-----------------------------------------|-----------------------|
| Content Service        | Notes, snippets, bookmarks, annotations | Content database      |
| Assessment Service     | Practice, tests, evaluation             | Question bank         |
| Analytics Service      | Performance metrics                     | Event stream          |
| Recommendation Service | AI suggestions                          | Analytics + Revision  |
| Goal Planner           | Daily plans                             | Analytics + Syllabus  |
| Revision Service       | Revision queue                          | Assessments + Content |
| Notification Service   | Reminders and alerts                    | Planner + Revision    |

## Non-Functional Requirements

| Category      | Requirement                | Target            |
|---------------|----------------------------|-------------------|
| Performance   | Dashboard load time        | < 2 seconds       |
| Availability  | Platform uptime            | > 99.5%           |
| Security      | Encrypted student data     | Mandatory         |
| Scalability   | Concurrent student support | Institution scale |
| Accessibility | Responsive web & mobile    | Required          |
| Backup        | Automated daily backups    | Required          |

## Security Requirements

- Role-based access control for Students, Parents, Teachers and Administrators.
- Encrypted storage of personal and assessment data.
- Secure authentication and session management.
- Audit logs for administrator actions.
- Compliance with institutional privacy requirements.

## Release Roadmap

| Phase         | Scope                                       | Outcome                  |
|---------------|---------------------------------------------|--------------------------|
| Phase 1 (MVP) | Dashboard, Learn, Practice, Chapter Tests   | Usable learning platform |
| Phase 2       | Rewards, Analytics, Syllabus, Saved Library | Improved engagement      |
| Phase 3       | Smart Revision & Goal Planner               | Personalized learning    |
| Phase 4       | AI Tutor, Parent & Teacher Apps             | Complete ecosystem       |

## Acceptance Criteria

- Students can complete an end-to-end learning cycle without assistance.
- All activities update analytics in near real time.
- Recommendations change according to student performance.
- Revision queues generate automatically after assessments.
- Goal Planner updates schedules when tasks are missed.

## Risks & Mitigation

| Risk                            | Impact            | Mitigation                              |
|---------------------------------|-------------------|-----------------------------------------|
| Low engagement                  | Reduced retention | Daily rewards and reminders             |
| Poor AI recommendations         | Student distrust  | Human-reviewed models and feedback loop |
| Large question bank duplication | Content quality   | Deduplication and tagging pipeline      |
| Performance bottlenecks         | Slow UX           | Caching and asynchronous processing     |

## Future Enhancements

Future releases will introduce an AI Tutor, conversational doubt solving, voice-assisted learning, adaptive learning paths, collaborative classrooms, parent mobile application, teacher workspace, predictive score estimation, multilingual content delivery and offline learning support.

## Final Conclusion

The Rizee Foundation module is designed as a complete AI-powered learning ecosystem rather than a conventional LMS. Every module—from learning and assessments to Smart Revision and Goal Planning—works together to create a personalized academic journey. This PRD provides the functional direction required for product, design and engineering teams to build a scalable and engaging platform.